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Explainable Ensemble Learning and Hybrid Feature Selection for Robust Low Birth Weight Prediction

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***Abstract*— Low Birth Weight (LBW) is a major public health issue characterized by a birth weight of less than 2500 grams. This study proposes a machine learning-based prediction of LBW using the Bangladesh Demographic and Health Survey (BDHS) 2017–2018 dataset, which contains 1863 maternal-infant records with 18 predictor features. Our approach addresses the critical gap between predictive power and clinical trust in this domain by introducing a methodologically robust framework that integrates an entire data preprocessing pipeline through a strict validation protocol to circumvent data leakage. Using a hybrid feature selection technique, a subset of 12 out of the 18 predictors was selected. We trained and evaluated 14 individual machine learning models, followed by the development of advanced voting ensembles using stratified k-fold cross-validation (CV). Our results show that a soft-voting ensemble of ExtraTrees and CatBoost establishes a new, robust performance benchmark, achieving a high score with an accuracy of 94.54% and a Receiver Operating Characteristic-Area Under the Curve (ROC-AUC) of 94.05%. Such high accuracy comes with a full model transparency through SHapley Additive exPlanations (SHAP) analysis. These results deliver a complete framework that successfully unifies high predictive accuracy with the methodological rigor and interpretability required for a trustworthy model in early LBW risk assessment.**

Keywords*— *Low Birth Weight, Machine Learning, Ensemble Methods, Feature Selection, Risk Assessment, Predictive Modeling, Maternal Health.

I. INTRODUCTION

Low birth weight (LBW), defined as a newborn weighing less than 2,500 grams, is a major health issue linked to short-term problems and long-term consequences like delayed development and increased neonatal mortality risk [1]. The WHO estimates 15% to 20% of all births are LBW, with prematurity as the leading cause in developed countries, and intrauterine growth restriction (IUGR) more common in developing countries [2]. The prevalence of LBW is higher in developing regions, including Bangladesh, with rates fluctuating from 17.7% in 2011 to 16% in 2017, and as high as 36% in some surveys [7]. Fig. 1 shows LBW prevalence across countries [3], [4]. Factors contributing to LBW risk include maternal health, gestational age and pregnancy nutrition [5], [6], [7]. LBW accounts for 40–60% of neonatal deaths and remains a key focus in global health agendas [8], [9]. Machine learning (ML) has been applied in healthcare for tasks like disease prediction [10], [11], but few studies predict LBW, especially in countries like Bangladesh. Most studies lack systematic feature selection and more advanced ML methods are needed to improve LBW prediction [13], [14].

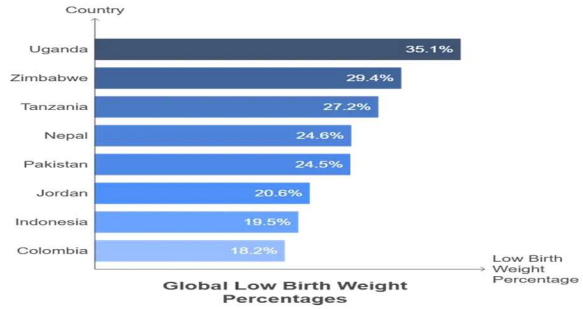


Fig. 1. Country-specific low birth weight percentages

Bangladesh Demographic and Health Survey (BDHS) 2017–2018 is an important source of information for such research, providing a wide-ranging, nationally representative dataset which is an essential opportunity to determine the maternal and child-related risk factors of LBW. Although previous attempts have been made using machine learning models on this dataset, particularly those using the BDHS 2017–2018 dataset, these are sometimes not methodologically sound. They often failed to adopt strict validation protocols to prevent data leakage and did not exploit the full predictive capacity of the most modern ensemble methods. To fill this gap, we propose

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in this paper a robust and reproducible framework achieving a new state-of-the-art on this crucial public health task. The following are the main contributions of this work:

- A new performance benchmark for LBW prediction in Bangladesh, which achieves an accuracy of 94.54% with a soft voting ensemble model, outperforming all prior reported results on this dataset.
- Identification of the clinically relevant LBW predictors and their association analysis by a hybrid feature selection method that could provide understandable information on important demographic and health determinants of LBW.
- A methodologically rigorous and transparent pipeline which accounts for data leakage and class imbalance appropriately, thereby providing a sound baseline for future research and potentially clinical risk stratification tools.

II. Literature Review

The prediction of low birth weight (LBW) and the detection of predisposing risk factors have been and are the current focus of global research as a result of its close association with neonatal morbidity and mortality. In recent years, machine learning (ML) methods have gained in popularity for LBW prediction. Initial studies aimed to identify clinical or demographic risk factors. For example, a study combined in 2014 noted non-smoking mothers, having greater pre-pregnancy weight, and no history of hypertension as relationships with less risk for LBW [6]. Subsequently, ML was employed for prediction models in later studies. Senthilkumar et al. [14] compared several ML techniques (Logistic Regression (LR), Naive Bayes (NB), Support Vector Machine (SVM), Neural Networks, and Decision Tree (DT)) for predicting LBW, and derived that Classification Tree models had the highest degree of predictive accuracy of 89.95%. An Indonesian study in 2018 compared Binary Logistic Regression and Random Forest (RF), which was dominated by RF and identified place of residence, wealth index and maternal characteristics as strong risk factors [15]. Another study on Bangladesh used ML, including LR and SVM and Random Forest (RF) to examine the effect of demographic characteristics on LBW [12]. Other research has also compared machine learning techniques. For instance, Loreto et al. (2019) evaluated six different algorithms on a Portuguese dataset and found that AdaBoost achieved the highest accuracy at 98%.[16]. In Iran, Random Forest and Logistic Regression are employed, and RF shows better accuracy (95% vs. 93%) [17]. A study by Yarlapati et al. (2017) utilized a Bayes minimum error rate classifier and achieved a high predictive accuracy [1]. Despite this discovery, analytical studies continue to point out important predictors, such as ethnicity and maternal adherence to iron and folic acid supplementation [5]. A careful analysis of this literature exposes several recurring deficiencies which hinder the reliability and clinical applicability of these models. Most studies give no robustness evaluation (e.g. cross-validation), which can introduce data leakage and potentially make strong performance claims in an unjustified manner. In addition, high-performing models are often treated as black boxes, limiting the exploration of feature importance and resulting in underreporting of clinical metrics, such as recall for the minority LBW class. This leaves a void: a demand for a methodologically responsible, interpretable, and clinically relevant reference standard. Our study directly tackles this gap in the current literature, with a highly rigorous framework to produce a reliable and accurate LBW prediction model.

III. Methodology

In this work, we utilize a machine learning pipeline that is both structured and methodologically sound, as illustrated in Fig. 2, to create a reliable prediction model for Low Birth Weight. The process begins with the BDHS 2017–18 dataset and initial data cleaning. A crucial early train-test split is performed to create an independent test set, ensuring all subsequent steps including hybrid feature selection, class imbalance handling, and model training are performed exclusively on the training data to prevent data leakage. Finally, a robust evaluation using stratified cross-validation and the untouched test set provides an unbiased assessment of model performance.

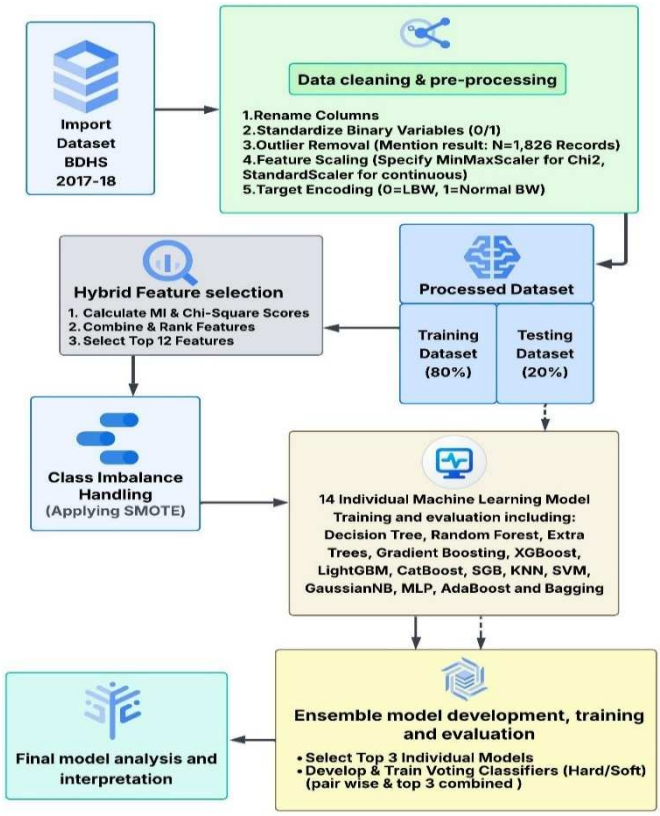


Fig. 2. Working Procedure of Proposed Low Birth Weight Prediction System.

A. Dataset Description

This study uses a maternal health dataset [20] with 1,863 records and 18 features, including maternal health, demographics and birth data. The response variable, y, indicates whether a baby was born with Low Birth Weight (1) or Normal Birth Weight (2). The dataset includes continuous (Weight, Height, Age at 1st sex), ordinal (Education level, Wealth index), categorical (Region, Place of Delivery) and binary (Child is twin, Sex of child) features. Initial analysis indicated the prevalence of data imbalance (1.85 mean target variable) as

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normal birth weight cases (Class 2) headed the count over the rest of data points. More details are provided in TABLE I.

TABLE I. DEFINING AND CATEGORIZING RESPONSE AND INDEPENDENT VARIABLES

Variable Name	Description	Categorization
Birth weight	Birth weight of newborn	1. Low Birth Weight (LBW) ii. Normal Birth Weight (NBW)
Age of the respondent	Mother's age in years	i. 15-19 ii. 20-24 iii. 25-29 iv. 30-34 v. 35-39
Highest education level	Mother's educational status	i. No education. ii. Primary iii. Secondary iv. Higher
Age of respondent at 1st birth	Mother's age at her first birth	i. 15-19 ii. 20-34 iii. 35-39
Age at 1st sex	Mother's age at first time of sex	i. <20 ii. 20-28 iii. >28
Wealth index	Wealth index status	(i) Poorest (ii) Poorer (iii) Middle (iv) Richer (v) Richest
BMI	Body Mass Index	(i) Underweight (ii) Normal weight (iii) Over weight iv. Obese
Height	Mother's height in meter	(No categorization provided)
Weight	Mother's weight in kg	(No categorization provided)
Place of delivery	Place of delivery of infant's	i. Home ii. Hospitals iii. Others
Educational attainment	Mother's educational level	i. No education ii. Incomplete primary iii. Complete primary iv. Incomplete secondary v. Complete secondary vi. Higher
Region	The administrative region where respondents reside	i. Dhaka ii.Barisal iii. Chittagong iv. Rangpur v. Rajshahi vi.Khulna vii. Mymensingh viii. Sylhet
Place of residence	Type of place of residence	i. Urban ii. Rural
Sex of child	Gender of the child	i. Female ii. Male
Child is twin	Single or multiple births	i. Single birth ii. 2nd of multiple
Child is alive	The child is alive	i. Yes ii. No
Birth order number	Order of the birth	(No categorization provided)
Delivery by caesarean section	Delivery in caesarean section	i. Yes ii. No
Antenatal care receive	Number of antenatal care receive	i. <=2 ii. >2

B. Dataset Cleaning and Preprocessing

A multi step data preprocessing pipeline was created for standardization, data quality improvements and machine learning optimizations. The preprocessing steps applied are as follow:

- Column Renaming: Column names were converted to lowercase and spaces were replaced with underscores (e.g., "Respondent Current age" to respondent_current_age).
- Binary Variable Standardization: Binary features (e.g. 1=Yes, 2=No) were re-coded to a 0/1 format. This included the target variable (y), which was transformed to 0=LBW, 1=Normal BW and other relevant features.
- Outlier Detection and Removal: Outliers were removed based on physiological ranges: height (<130 cm or >190 cm) and weight (<35 kg or >90 kg), reducing the dataset to 1,826 rows after removing 37 records.
- Train-Test Split: The preprocessed dataset was then split into a training set (80%, 1460 rows) and an independent test set (20%, 366 rows) using train_test_split with random_state=42 for reproducibility. The test set was held out from this point forward.
- Feature Scaling: As continuous variables (weight, height) were scaled with StandardScaler. Only the training data was used to fit the scaler and then both the training and test sets were transformed using the fitted scaler. For Chi-Square feature selection, numeric attributes were likewise scaled via min-max in a fit-on-train, transform-both manner.
- Target Variable Encoding: The binary target variable y in the training and test sets was encoded with LabelEncoder for compatibility with library functions.
- Feature Engineering: No new features were created, as the dataset already contained useful variables like BMI and maternal age. Categorical features like region and wealth_index were handled directly by the tree-based models.

C. Hybrid Feature Selection

To identify the most relevant predictors and improve model interpretability, a hybrid feature selection approach was implemented exclusively on the 80% training set:

- Mutual Information (MI): MI score between each feature and the target (y) was computed, using mutual_info_classif; it measures simultaneously linear and non-linear relationship. The features were ordered based on their MI scores.
- Chi-Square Test (χ^2): SelectKBest with chi^2 computed the chi-squared statistic of each feature, assessing independence for categorical or non-negative features. Features were ranked by their χ^2 scores after applying MinMaxScaler.
- Hybrid Ranking: MI and χ^2 rankings were combined to leverage the strengths of both methods.
- Feature Selection: The top 12 features were selected based on combined importance scores. The most influential feature was age_at_1st_sex (14.0), followed by region (13.5) and birth_order_number (13.0). Other key predictors included type_of_place_of_residence

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(12.5), highest_education_level (12.0) and child_is_alive (11.0). Features like respondent_current_age (3.0) and Body Mass Index (BMI) (4.0) had lower scores and were excluded. A visual representation of the importance scores is shown in Fig. 3.

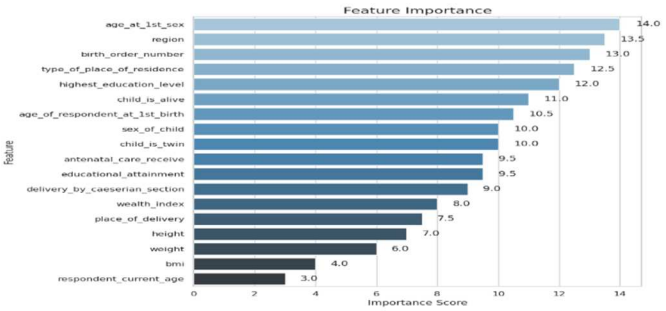


Fig. 3. Feature Importance for LBW Prediction

D. Class Imbalance Handling

The training data suffered from extreme class imbalance, with just 209 samples for the minority LBW class (Class 0), against 1,251 for the Normal BW class (Class 1). The Synthetic Minority Oversampling Technique (SMOTE) was performed on training data in order to avoid the learning algorithms being biased towards the majority class. This technique was limited to the training data and generates synthetic instances of the minority class. This led to a completely balanced training dataset, where the LBW class was upsampled to 1,251 samples, as shown in Fig. 4.

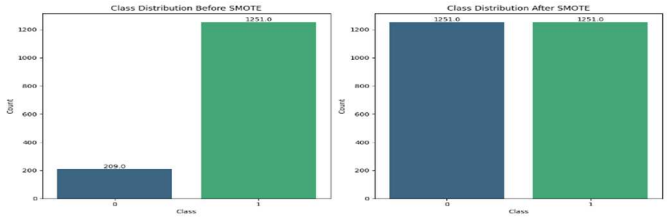


Fig. 4. Class Distribution Before and After SMOTE

E. Model Training and Evaluation

1. *Environmental Setup: The experiment was conducted on a personal computer with 16 GB RAM, Core i7 12th Gen CPU, NVIDIA GeForce RTX 3050 GPU, and 512 GB disk space, using Google Co-Lab, Python tools, and other libraries, as shown in TABLE I-III.*

2. Model Selection: To perform a robust model selection process and to get reliable estimate of performance, we used a stratified 5-fold CV strategy. This was only conducted on the SMOTE-balanced and 12-feature-selection-applied processed training data. This CV process was utilized in training and validating a suite of 14 applied machine learning algorithms. The average performance across the five folds was used to select the top performing models. These selected models were then retrained on the entire training set and their final generalization performance was assessed a single time on the untouched test set.

3. Advanced Ensemble Techniques – Voting Classifier: To boost the performance of the individual models, we built voting ensembles using the top three classifiers, found in our first evaluation. Different Hard and Soft Voting setups were generated, and each was evaluated through stratified 5-fold cross-validation on the training set, to robustly choose the best one. Lastly, the ensemble with the highest performance during this process was re-trained on the complete training set for a final unbiased assessment on the unexamined test set.

4. Evaluation Metrics: Final model performance was evaluated on the independent test set with a broad set of metrics. We computed the overall accuracy, precision, F1–Score and ROC-AUC . We placed special emphasis on examining the metrics of the minority (LBW) class since it is an important clinical consideration. This was followed by a breakdown of the confusion matrix, to derive LBW-specific recall (sensitivity) and LBW-specific precision, as a means to ensuring that this tool could function as an effective risk-screening tool.

IV. Results and Discussion

This section presents the empirical outcomes from the application of various machine learning models to the Low Birth Weight (LBW) prediction task. We used standard metrics—Accuracy, Precision, Recall, F1-Score and Receiver Operating Characteristic-Area Under the Curve (ROC-AUC) on a separate unseen test set to measure generalization capability.

A. Individual Model Performance

The performance of 14 models on the unseen test set is detailed in TABLE II. Tree-based ensembles clearly dominated, with CatBoost achieving the highest accuracy at 92.01% and an F1-Score of 92.51%. ExtraTrees (90.42% Accuracy) and LightGBM (89.62% Accuracy) also performed strongly. In contrast, models like Gaussian Naive Bayes (55.11%) were significantly outperformed. While these results establish a strong baseline, their high recall is for the majority class, motivating the use of advanced ensembles to improve the critical minority LBW class prediction.

TABLE II. Top Individual Models Evaluation Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
CatBoost	92.01%	86.99%	98.8%	92.51%	92.02%
ExtraTrees	90.42%	89.23%	92.06%	90.63%	94.07%
LightGBM	89.62%	83.97%	98.10%	90.48%	92.66%
XGBoost	88.82%	84.12%	95.87%	89.61%	92.57%
GradientBoosting	88.66%	81.77%	99.68%	89.84%	92.05%
SGB	88.66%	81.77%	99.68%	89.84%	92.05%
RandomForest	88.34%	84.57%	93.97%	89.02%	93.71%
Bagging	85.94%	85.14%	87.30%	86.21%	91.93%
DecisionTree	80.83%	81.97%	79.37%	80.65%	81.07%
AdaBoost	78.43%	76.63%	82.22%	79.33%	85.48%

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MLP	77.96%	79.80%	75.24%	77.45%	85.02%
KNN	76.52%	88.53%	61.27%	72.42%	85.27%
SVM	67.41%	71.10%	59.37%	64.71%	73.30%
GaussianNB	55.11%	52.89%	98.73%	68.88%	60.77%

B. Voting-Based Ensemble Performance

To enhance predictive performance, voting ensembles were constructed using the top-performing individual models. As specified in TABLE III, the best configuration turned out to be the soft-voting ensemble of ExtraTrees and CatBoost, with an overall accuracy of 94.54% and F1-Score of 96.76%. While the table reports a high precision (98.36%) and recall (95.22%), these metrics reflect the model's success in identifying the majority 'Normal Birth Weight' (NBW) class. For a true assessment of clinical utility, we must analyze the model's performance on the minority 'Low Birth Weight' (LBW) class using the confusion matrix (Fig. 5). While the model records LBW Precision of 75.81%. However, the model's primary strength is highlighted by its impressive LBW Recall (Sensitivity) of 90.38%. By correctly identifying 47 out of 52 LBW infants, the system proves its robustness in detecting the most critical high-risk cases. The accuracy across the 5-fold cross-validation also confirmed a stable performance corresponding to a low standard deviation (0.0104), thereby validating the generalizability of the model (TABLE IV).

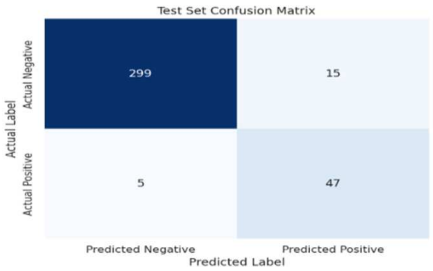


Fig. 5. Confusion Matrix of Pairwise Soft Voting Classifier (ExtraTrees - CatBoost)

TABLE III. TOP VOTING CLASSIFIERS INCLUDING PAIRWISE AND FULL ENSEMBLE RESULTS

Voting Model	Voting Type	Accuracy	Precision	Recall	F1 Score	ROC-AUC
ExtraTrees - CatBoost	Soft	94.54%	98.36%	95.22%	96.76%	94.05%
ExtraTrees - LightGBM	Soft	91.05%	86.89%	96.83%	91.59%	95.52%
ExtraTrees - CatBoost	Hard	91.05%	90.34%	92.06%	91.19%	-
ExtraTrees - LightGBM	Hard	90.58%	89.75%	91.75%	90.74%	-
Top 3	Soft	90.26%	85.28%	97.46%	90.96%	95.28%
LightGBM - CatBoost	Soft	89.78%	84.01%	98.41%	90.64%	93.18%
Top 3	Hard	89.78%	84.01%	98.41%	90.64%	-
LightGBM - CatBoost	Hard	89.78%	84.20%	98.10%	90.62%	-

TABLE IV. METRICS ACROSS 5 FOLDS FOR THE EXTRATREES-CATBOOST SOFT VOTING ENSEMBLE MODEL

	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Fold 1	0.9460	0.9750	0.9580	0.9660	0.9410
Fold 2	0.9420	0.9810	0.9500	0.9650	0.9420
Fold 3	0.9490	0.9870	0.9540	0.9700	0.9450
Fold 4	0.9400	0.9790	0.9490	0.9630	0.9400
Fold 5	0.9440	0.9850	0.9510	0.9670	0.9430
Mean	0.9442	0.9816	0.9520	0.9662	0.9422
Standard Deviation	0.0104	0.0090	0.0075	0.0085	0.0104
95% Confidence Interval	±0.0144	±0.0130	±0.0110	±0.0120	±0.0144

To interpret the model's predictions, we used SHapley Additive exPlanations (SHAP). The summary plot (Fig. 6) identifies the most significant predictors are geographic region, maternal age at first sex and birth order number. The plot reveals clinically intuitive relationships; for example, a lower age_at_1st_sex (younger maternal age) consistently increases the risk of an LBW prediction. Such analysis offers much needed transparency and proves the clinical relevance of the model by confirming its decisions are driven by factors well-established in maternal health. However, the delineation of these top predictors is in agreement with other public health studies in developing countries, which has repeatedly highlighted maternal age and sociodemographic factors as the most important risk factors for LBW [4],[7]. This kind of transparency is important, it gives an explicit clinical rationale for the model's utility in decision support.

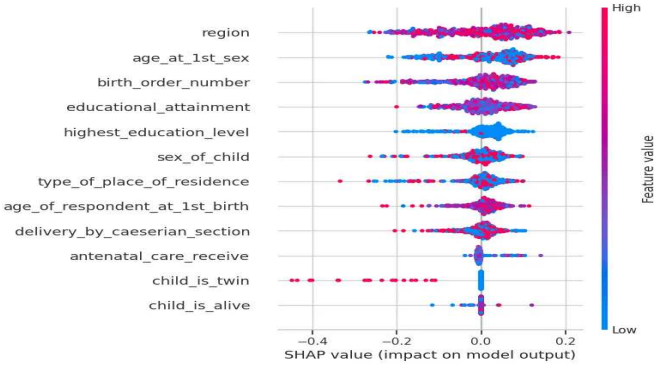


Fig. 6. SHAP Summary: ExtraTrees-CatBoost Feature Importance

C. Comparison with Existing Work :

To provide a sense of scale of our results, we compared our proposed ensemble model against recent work on LBW prediction, as reported in TABLE V. Two studies , Reza et al. [18] and Pollob et al. [3] were based on the same BDHS 2017–18 dataset. Reza et al. applied several classifiers (RF, DT, SVM, NB, XGBoost, AdaBoost) with a best accuracy of 87.30%. Pollob et al. [3] achieved 87.60% using Logistic Regression. In contrast, our ensemble of ExtraTrees - CatBoost, with the use of preprocessing, SMOTE and hybrid feature selecion, reaches 94.54% accuracy, showing a huge enhancement. Other works

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used different datasets. Senthilkumar D et al. [14] obtained 89.90% with the DT model on hospital data, but DTs are prone to overfitting and are not powerful for robust ensembles. Ranjbar et al. [19] employed XGBoost on Iranian IMaN Net data limiting around 79% accuracy, probably due to small data size and dataset's features. Overall, our ensemble stably outperformed models applied to the same dataset and proved competitive across others, which reinforces the crucial steps of robust preprocessing, feature selection and ensemble design to achieve high LBW accuracy.

TABLE V. COMPARISON OF MACHINE LEARNING MODELS FOR EARLY LBW DETECTION ACROSS DIFFERENT STUDIES

Authors	Dataset	Method	Accuracy
Tasneem Binte Reza et al. [18]	BDHS 2017–2018 survey data	RF,DT,SVM,NB,XGBoost, AdaBoost	87.30%
S. M. A. Islam Pollob et al.[3]	BDHS 2017–2018 survey data	LR	87.60%
Amene Ranjbar et al.[19]	Iranian Maternal and Neonatal Network (IMaN Net)	XGBoost	79%
Senthilkumar D et al.[14]	Secondary data from a medical center	DT	89.90%
Proposed Model	BDHS 2017–2018 survey data	Ensemble (ExtraTrees - CatBoost)	94.54%

V. CONCLUSION AND FUTURE WORK

This paper successfully developed and validated a high-performance ML framework for early LBW risk stratification, achieving a new, robust performance benchmark on the BDHS 2017-18 dataset. The main contribution of our work is in showing a complete framework integrating strong predictive performance with the procedural rigour and clinical interpretability necessary to yield a credible model. The key findings of this work are threefold. First, we obtained a better predictive accuracy of 94.54% with our soft-voting ensemble of ExtraTrees and CatBoost. Second, this performance is supported by a methodologically robust pipeline including a hybrid feature selection strategy to identify the most predictive features and rigorous validation processes using stratified k-fold cross-validation and an independent test set to ensure stability and reduce overfitting. Third, the use of SHAP analysis allowed us to go beyond the black box model, providing interpretable decision aids that connected predictions to known clinical risk factors, thereby providing trust in the model and making it appropriate for decision support. This framework builds a strong baseline, however, the next step should be external validation on various, contemporary, prospective datasets to allow one to advocate for its generalizability. Future work will also focus on using LLMs and gain value from the unstructured clinical data. Additionally, evaluation of model calibration and net clinical benefit will be central to potentially implement this work into a true clinical decision-making tool.

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